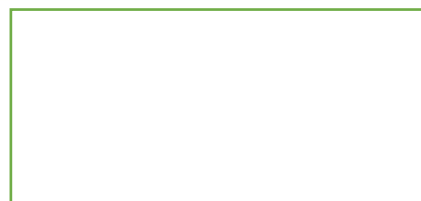


**TRANSCRIPTOMIC ANALYSIS OF
TEMEPHOS - RESISTANCE IN *AEDES
AEGYPTI* POPULATIONS FROM
COLOMBIA.**



INTRODUCTION



Figure 1: *Aedes aegypti* (Exebio-Chepe et al., 2024).

Aedes aegypti is a mosquito species of major global public health concern and is widely recognised as the primary vector of several human arboviral diseases. Its effectiveness as a vector is linked to its close association with human environments, preference for human blood meals, and ability to exploit artificial water containers for breeding (Grisales *et al.*, 2013). These characteristics have allowed *Ae. aegypti* to establish stable populations in urban and peri-urban settings, particularly in tropical and subtropical regions.

Diseases transmitted by *Ae. aegypti* include dengue, Zika, chikungunya and yellow fever. Dengue fever represents the greatest burden, with an estimated 390 million infections annually worldwide (Davila-Barboza et al. 2024). Chikungunya infection is associated with chronic musculoskeletal symptoms, while Zika virus outbreaks have been linked to severe congenital abnormalities, highlighting the medical and socio-economic impact of these infections (Gan et al. 2021).

Historically, *Ae. aegypti* originated in Africa, where ancestral populations bred in forest habitats. The species spread globally through human movement and trade, particularly during the transatlantic shipping period, and later expanded rapidly with urbanisation and increased water storage practices (Poulton et al. 2024)). As a result, many regions now face persistent mosquito infestations and recurring disease outbreaks.

Mosquito control strategies have evolved over time. Early approaches focused on environmental management and larval habitat reduction. The introduction of chemical insecticides such as DDT in the mid-20th century led to major reductions in mosquito populations; however, concerns over environmental persistence and toxicity resulted in its restriction. Subsequently, organophosphates and pyrethroids became the primary chemical tools for vector control (Hemingway *et al.*, 2016). Among these, temephos has been widely used as a larvicide in domestic water containers due to its effectiveness and affordability (WHO, 2011). Prolonged reliance on chemical control has led to the emergence of insecticide resistance, threatening the sustainability of current control programmes.

Modern molecular approaches, including RNA sequencing (RNA-seq), enable genome-wide investigation of gene expression changes associated with resistance. This study applies RNA-seq to identify differentially expressed genes linked to temephos resistance in *Ae. aegypti*, providing insights relevant to future vector control strategies.



Figure 2: Global distribution of the *Aedes aegypti* mosquito including historical records. Presence (red dots) (Kraemer et al.,2015b)

Biology of *Aedes aegypti*

Aedes aegypti is a mosquito from the family Culicidae and the genus Aedes. It has a black body with white lyre-shaped markings on the thorax and white-banded legs, which make it easy to distinguish from other mosquitoes like *Culex*. Besides these physical features, *Ae. aegypti* is known for biting during the day and preferring humans as hosts. These behaviours make it an effective carrier of diseases.

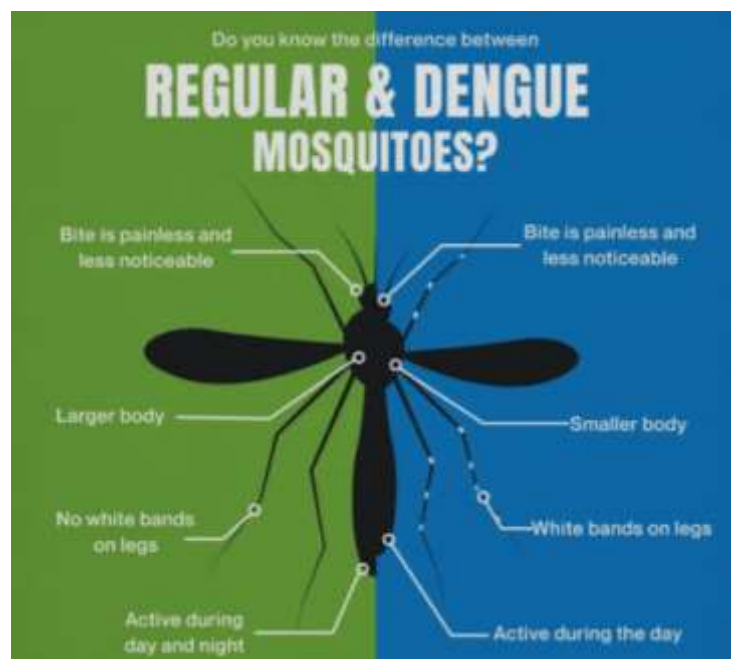


Figure 3: Differences between regular and dengue mosquitoes (Piedra et al., 2024).

The life cycle of *Ae. aegypti* has four stages: egg, larva, pupa, and adult. Females lay eggs individually on moist surfaces just above the water. The eggs can survive dry conditions for several months, allowing populations to persist even when water is scarce. When eggs are flooded, they hatch into larvae, which pass through four instar stages and feed on microorganisms and organic matter. This feeding makes larvae active and sensitive to larvicides. After the larval stage, they develop into pupae, a short, non-feeding stage where metamorphosis occurs. Adult mosquitoes emerge from pupae at the water surface. Female adults need blood to produce eggs and are the main agents of disease transmission. Larvae are targeted in control programs because they cannot move away, and controlling them prevents adults from emerging and spreading disease. (Turner *et al.*, 2025)

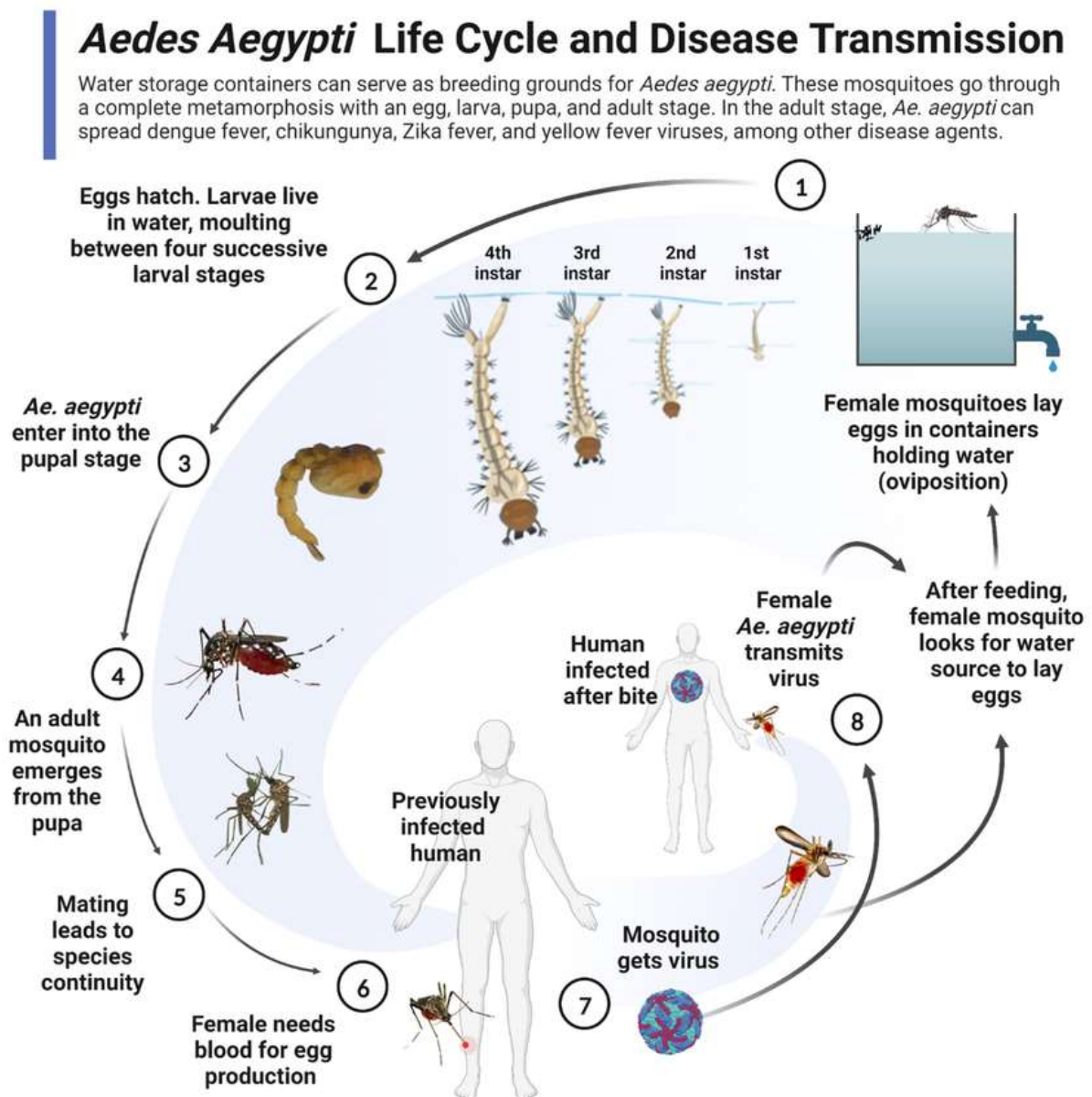


Figure 4: The life cycle of *Ae. aegypti* and disease transmission (Turner *et al.*, 2025)

Vector Control and Insecticide Resistance

Mosquito control can target either larvae or adults. Larval control methods, such as larviciding or eliminating breeding sites, prevent mosquitoes from reaching adulthood and are more sustainable. Adult control, like aerial insecticide spraying, provides short-term results and can quickly lead to resistance.

Temephos is a common larvicide used against *Ae. aegypti*, working by inhibiting acetylcholinesterase, an enzyme essential for normal nerve function. This overstimulation of nerves causes paralysis and death of larvae. Mosquitoes can develop resistance through various mechanisms. Metabolic resistance occurs when detoxifying enzymes like cytochrome P450 and esterases become overactive. Target-site resistance arises from mutations that prevent the insecticide from binding properly. Additionally, changes in gene regulation can enhance survival by improving detoxification and stress responses (Martínez-Mercado et al., 2022).

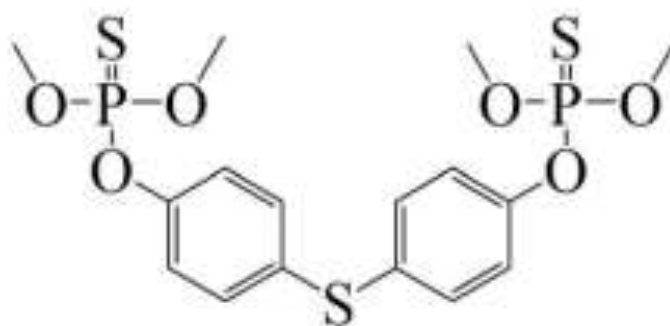


Figure 5: Chemical structure of Temephos (MDPI, 2019)

Prolonged and widespread use of temephos has led to the emergence of resistance in *Aedes aegypti* populations, reducing its effectiveness. High resistance levels have been reported in various regions, including Havana, Cuba, where resistant populations persisted even after temephos application ceased (Piedra et al., 2024). Insecticide resistance is often linked to genetic and transcriptomic changes, such as altered expression of detoxification enzymes and changes in target-site sensitivity. RNA sequencing (RNA-seq) offers a powerful method to identify differentially expressed genes associated with resistance. This study uses RNA-seq to compare temephos-susceptible and resistant *Aedes aegypti* populations from Colombia, aiming to uncover the molecular mechanisms underlying resistance (Piedra et al., 2024).

Experimental Design Overview

Two *Ae. aegypti* populations were studied: the temephos-susceptible Bello population and the resistant Cucuta population. The WHO larvicide assay classified these populations for gene expression comparison. Four biological replicates were analyzed for each population, with pooled RNA from 30 larvae per replicate to reduce individual variability and enhance statistical power.

METHODOLOGY

Mosquito populations and sample preparation

Two *Aedes aegypti* populations from Colombia were analyzed: the temephos-susceptible Bello population and the resistant Cucuta population, classified based on WHO larvicide bioassays (WHO, 2005). Fourth-instar larvae, which are directly targeted by larvicides and exhibit active metabolic processes (Clements, 1992), were used. To reduce individual variability and obtain representative gene expression, larvae were pooled for RNA extraction. Four biological replicates per population were prepared, each consisting of 30 larvae, which were rinsed to remove debris and immediately processed to preserve RNA integrity.

RNA extraction and quality control

Total RNA was extracted from pooled larval samples using standard protocols designed to isolate high-quality RNA. Cellular membranes were lysed to release nucleic acids, and contaminants like proteins and lipids were removed. RNA samples were treated to eliminate genomic DNA contamination, ensuring that analyses reflected transcript abundance, not genomic sequences. RNA quality and integrity were assessed before sequencing, following established quality control procedures to ensure accuracy in transcriptomic analysis (Conesa et al., 2016). Only samples meeting quality standards proceeded to library preparation.

Library preparation and Illumina sequencing

mRNA was enriched from total RNA to focus on protein-coding genes potentially involved in insecticide resistance. The enriched mRNA was fragmented, reverse-transcribed to cDNA, and then second-strand synthesis was performed to generate double-stranded cDNA. Sequencing adapters were ligated to enable amplification and recognition by the sequencing platform, following the QIAseq™ Stranded mRNA Selected Kit Handbook. Stranded library preparation preserved transcript directionality for accurate genome mapping. Libraries were amplified and sequenced on the Illumina NovaSeq 6000 platform, utilizing sequencing-by-synthesis technology. Fluorescently labeled nucleotides were incorporated into growing DNA strands, and emitted signals were recorded to generate nucleotide sequences (Illumina, 2020). The output consisted of gene-level read counts representing transcript abundance for each replicate, provided as an Excel dataset for bioinformatic analysis.

Blot preparation and gene identification (validation concept)

Although RNA sequencing provides genome-wide expression data, traditional blot techniques like Northern blotting have been used historically to validate transcript presence and size. This involves separating RNA fragments by size through gel electrophoresis, transferring them to a membrane, and hybridizing with labeled probes complementary to target genes (Sambrook & Russell, 2001). While blot validation was not performed in this study, understanding this method highlights how RNA-seq replaces blot-based techniques by offering high-resolution, quantitative transcript detection across the genome.

Meta-table construction and dataset standardization

Raw gene count data from all biological replicates were imported into Microsoft Excel to create a meta-table for statistical analysis. Column headers were standardized by removing spaces and special characters, replacing them with underscores, ensuring compatibility with R and other analysis software. Genes with zero counts across all replicates in both populations were removed to reduce noise and improve statistical robustness. Mean expression values were calculated for the susceptible (Bello) and resistant (Cucuta) populations using the four replicates. This reduces variability and highlights consistent population differences. Additional columns were added for fold change values, p-values, and transformed statistical values. This meta-table served as the primary dataset for downstream analysis and visualization.

Differential gene expression analysis

Differential gene expression analysis was performed to identify genes with significant expression differences between the temephos-resistant and susceptible populations. Fold change values were calculated by comparing expression levels in the resistant population to the susceptible one. Genes with a fold change greater than two ($\log_2 \text{FC} > 1$) were considered biologically meaningful. Statistical significance was assessed using a two-tailed Student's t-test with a significance threshold of $p < 0.05$. Fold change values were $\log_2$ -transformed, and p-values were transformed to $-\log_{10}(p)$ to stabilize variance and support graphical interpretation (Love et al., 2014).

Visualisation of differential gene expression

Visualization methods helped interpret differential expression results. Volcano plots were generated using [R Studio](#), with $\log_2$ fold change on the x-axis and $-\log_{10}(\text{p-value})$ on the y-axis, enabling quick identification of genes that were both highly differentially expressed and statistically significant. Heatmaps were created for the top differentially expressed genes based on fold change and statistical significance, displaying normalized expression values across all replicates for direct comparison of expression patterns between the susceptible and resistant populations.

Selection of top 3 genes for downstream analysis

Based on volcano plot and heatmap results, the three most highly overexpressed genes in the resistant Cucuta population were selected for downstream bioinformatic analysis. Selection prioritised genes exhibiting both large fold changes and strong statistical significance, increasing the likelihood that these genes contribute to temephos resistance mechanisms.

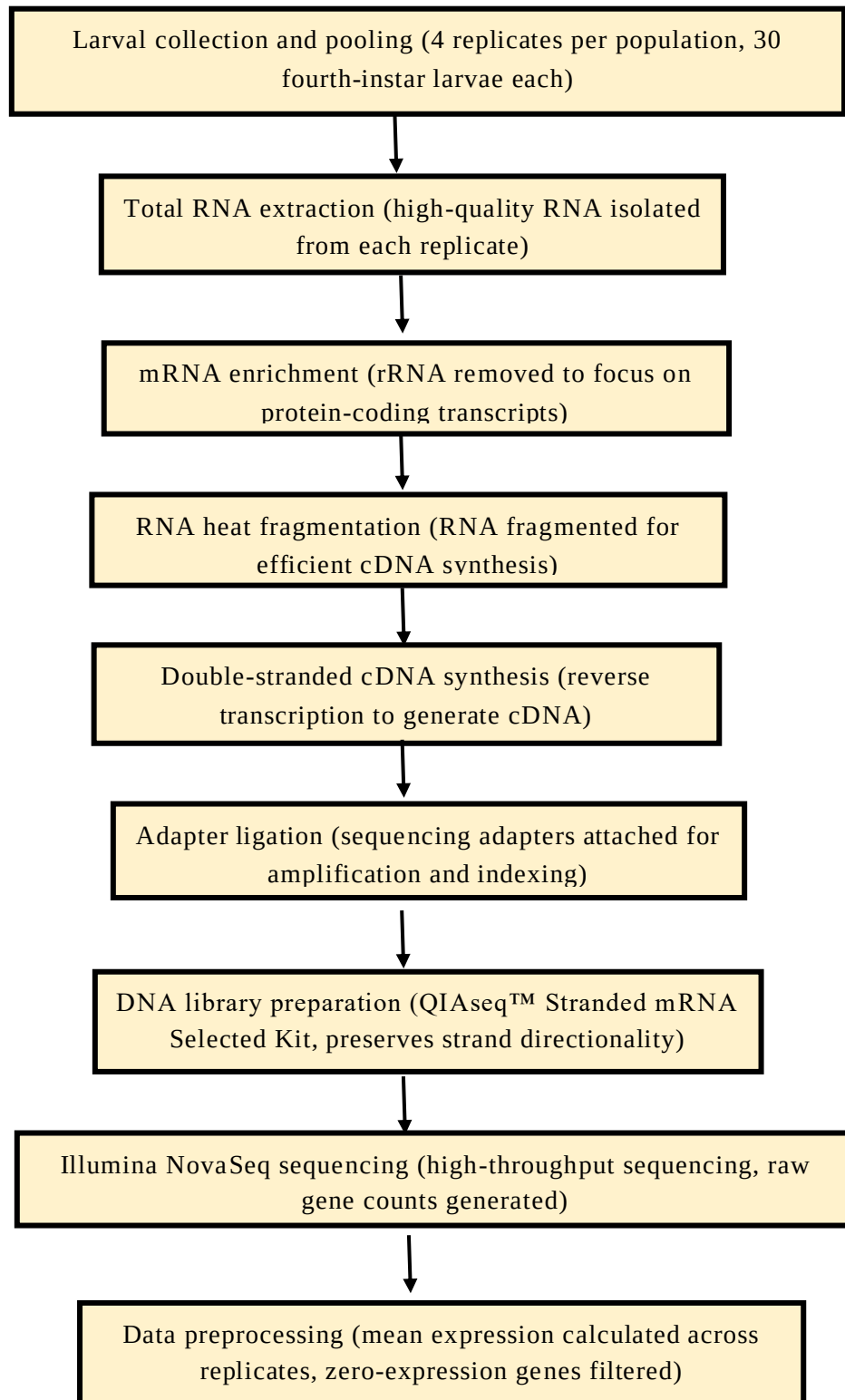
Functional annotation and bioinformatics analysis

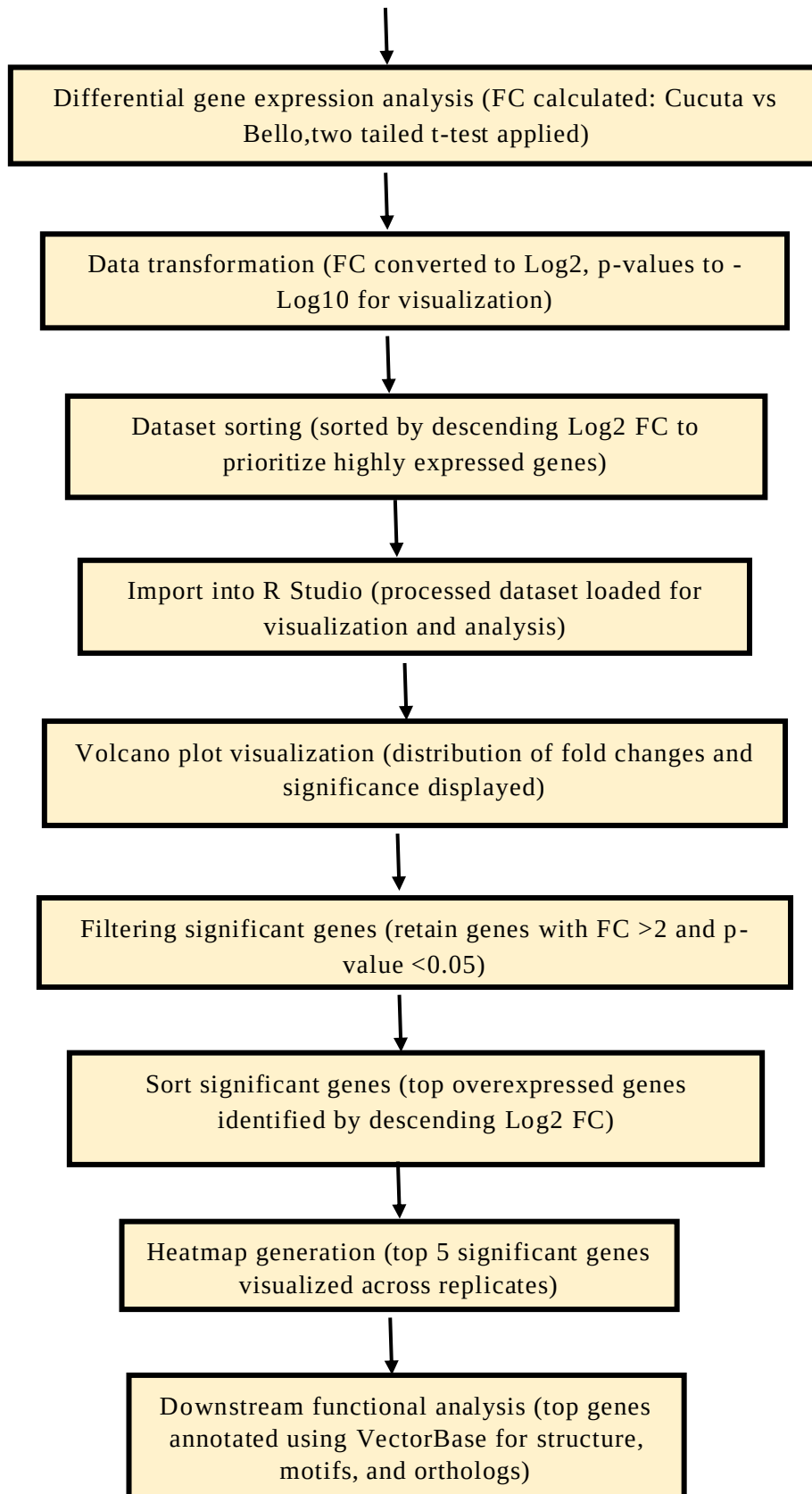
Functional annotation of selected genes was performed using [VectorBase](#), a specialized database for invertebrate disease vectors. Gene structure, including exon–intron organization, gene length, and genomic DNA size, was analyzed to identify potential regulatory features. Chromosomal locations and genomic coordinates were determined to assess the genomic context and possible gene clustering. Predicted protein sequences were examined for conserved domains and functional motifs. Orthologous genes in other mosquito species were identified to evaluate evolutionary conservation, and functional predictions were supported by sequence homology and published literature on insecticide resistance.

Bioinformatic pipeline and flow-chart description

The bioinformatic pipeline followed a structured and reproducible workflow comprising the following steps:

PIPELINE:





Pipeline validation and data reliability

To ensure reliability and reproducibility, data quality was assessed prior to downstream processing, and biological replicates were used for each mosquito population. This reduced the influence of individual variability and increased statistical confidence in observed expression differences (Conesa et al., 2016).

Gene expression comparisons were conducted only after data cleaning and filtering, ensuring that observed differences reflected biological variation rather than technical bias. The structured and reproducible pipeline enabled robust analysis and allowed replication of the workflow in future transcriptomic studies.

RESULTS

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Gene_VectorBase_identifier	Bello_1	Bello_2	Bello_3	Bello_4	Cucuta_1	Cucuta_2	Cucuta_3	Cucuta_4	Mean_Bello	Mean_Cucuta		
2	AAEL000001	570	764	537	537	415	521	532	532	602	500		
3	AAEL000004	803	758	873	873	769	936	711	711	826.75	781.75		
4	AAEL000005	124	104	82	82	63	93	71	71	98	74.5		
5	AAEL000006	8	6	6	6	2	4	3	3	6.5	3		
6	AAEL000008	8	13	14	14	8	20	8	8	12.25	11		
7	AAEL000010	1	3	2	2	2	0	0	0	2	0.5		
8	AAEL000011	0	0	0	0	0	0	0	0	0	0		
9	AAEL000012	4	2	2	2	3	1	3	3	2.5	2.5		
10	AAEL000013	302	391	344	344	340	411	424	424	345.25	399.75		
11	AAEL000016	474	652	437	437	508	670	643	643	500	616		
12	AAEL000017	0	0	0	0	0	0	0	0	0	0		
13	AAEL000018	0	0	0	0	0	0	0	0	0	0		
14	AAEL000020	78	81	65	65	87	95	61	61	72.25	76		
15	AAEL000021	551	402	384	384	206	319	409	409	430.25	335.75		
16	AAEL000022	1269	1308	1268	1268	858	1068	940	940	1278.25	951.5		
17	AAEL000024	181	291	162	162	152	188	176	176	199	173		
18	AAEL000025	3947	3050	2186	2186	2579	1987	3304	3304	2842.25	2793.5		
19	AAEL000026	143	102	115	115	74	95	91	91	118.75	87.75		
20	AAEL000027	0	1	1	1	1	4	4	4	0.75	3.25		
21	AAEL000028	594	873	874	874	792	706	482	482	803.75	615.5		
22	AAEL000029	653	781	630	630	661	1021	549	549	673.5	695		

Figure 6: Meta data sheet [A value of 0 indicates no detectable gene expression (zero read counts) in that sample.]

	A	B	C	D	E	F	G	H	I	J	K	L
1	Gene_VectorBase_identifier	Bello_1	Bello_2	Bello_3	Bello_4	Cucuta_1	Cucuta_2	Cucuta_3	Cucuta_4	Mean_Bello	Mean_Cucuta	FC
2	AAEL000001	570	764	537	537	415	521	532	532	602	500	
3	AAEL000004	803	758	873	873	769	936	711	711	826.75	781.75	
4	AAEL000005	124	104	82	82	63	93	71	71	98	74.5	
5	AAEL000006	8	6	6	6	2	4	3	3	6.5	3	
6	AAEL000008	8	13	14	14	8	20	8	8	12.25	11	
7	AAEL000010	1	3	2	2	2	0	0	0	2	0.5	
9	AAEL000012	4	2	2	2	3	1	3	3	2.5	2.5	
10	AAEL000013	302	391	344	344	340	411	424	424	345.25	399.75	
11	AAEL000016	474	652	437	437	508	670	643	643	500	616	
14	AAEL000020	78	81	65	65	87	95	61	61	72.25	76	
15	AAEL000021	551	402	384	384	206	319	409	409	430.25	335.75	
16	AAEL000022	1269	1308	1268	1268	858	1068	940	940	1278.25	951.5	
17	AAEL000024	181	291	162	162	152	188	176	176	199	173	
18	AAEL000025	3947	3050	2186	2186	2579	1987	3304	3304	2842.25	2793.5	
19	AAEL000026	143	102	115	115	74	95	91	91	118.75	87.75	
20	AAEL000027	0	1	1	1	1	4	4	4	0.75	3.25	
21	AAEL000028	594	873	874	874	792	706	482	482	803.75	615.5	
22	AAEL000029	653	781	630	630	661	1021	549	549	673.5	695	
23	AAEL000032	191231	215448	166335	166335	142449	187434	200213	200213	184837.25	182577.25	
24	AAEL000033	89	113	83	83	82	88	90	90	92	87.5	
25	AAEL000035	1	4	2	2	1	3	0	0	2.25	1	

Figure 7: Meta data sheet After removing 0.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Gene_VectorBase_identifier	Bello_1	Bello_2	Bello_3	Bello_4	Cucuta_1	Cucuta_2	Cucuta_3	Cucuta_4	Mean_Bello	Mean_Cucuta	FC	
2	AAEL026689	2	2	0	0	2446	3125	6	6	1	1395.75	1395.75	
3	AAEL020127	2	2	3	3	4652	6699	24	24	2.5	2849.75	1139.9	
4	AAEL007951	0	1	0	0	99	100	155	155	0.25	127.25	509	
5	AAEL013757	0	2	0	0	13	59	171	171	0.5	103.5	207	
6	AAEL005656	0	2	1	1	5	673	6	6	1	172.5	172.5	
7	AAEL013031	166	115	57	57	31240	36217	57	57	98.75	16892.75	171.0658	
9	AAEL014893	0	1	0	0	56	4	41	41	0.25	35.5	142	
10	AAEL026896	2	2	7	7	1021	1107	16	16	4.5	540	120	
11	AAEL017547	1	0	0	0	43	17	19	19	0.25	24.5	98	
14	AAEL005656	2	2	0	0	1	326	0	0	1	81.75	81.75	
15	AAEL011642	34	77	235	235	18554	27057	123	123	145.25	11464.25	78.92771	
16	AAEL009790	20	58	177	177	14607	18748	163	163	108	8420.25	77.96528	
17	AAEL002246	1	1	0	0	46	39	35	35	0.5	38.75	77.5	
18	AAEL027068	2	1	1	1	3	2	173	173	1.25	87.75	70.2	
19	AAEL012852	4	5	3	3	374	623	7	7	3.75	252.75	67.4	
20	AAEL014617	2	0	1	1	54	90	62	62	1	67	67	
21	AAEL005656	9	8	13	13	3	2791	10	10	10.75	703.5	65.44186	
22	AAEL007542	80	72	76	76	7406	10240	43	43	76	4433	58.32895	
23	AAEL004590	1	1	2	2	151	194	0	0	1.5	86.25	57.5	
24	AAEL022419	0	0	1	1	70	7	18	18	0.5	28.25	56.5	
25	AAEL023843	4	3	0	0	83	109	101	101	1.75	98.5	56.28571	

Figure 8: Fold change calculation and sorting

O11																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N			
1	Gene_VectorBase_identifier	Bello_1	Bello_2	Bello_3	Bello_4	Cucuta_1	Cucuta_2	Cucuta_3	Cucuta_4	Mean_Be_T	Mean_Cuc_T	FC	logFC				
2	AAEL026689	2	2	0	0	2446	3125	6	6	1	1395.75	1395.75	10.44682				
3	AAEL020127	2	2	3	3	4652	6699	24	24	2.5	2849.75	1139.9	10.15469				
4	AAEL007951	0	1	0	0	99	100	155	155	0.25	127.25	509	8.991522				
5	AAEL013757	0	2	0	0	13	59	171	171	0.5	103.5	207	7.693487				
6	AAEL005656	0	2	1	1	5	673	6	6	1	172.5	172.5	7.430453				
7	AAEL013031	166	115	57	57	31240	36217	57	57	98.75	16892.75	171.0658	7.418408				
9	AAEL014893	0	1	0	0	56	4	41	41	0.25	35.5	142	7.149747				
10	AAEL026896	2	2	7	7	1021	1107	16	16	4.5	540	120	6.906891				
11	AAEL017547	1	0	0	0	43	17	19	19	0.25	24.5	98	6.61471				
14	AAEL005656	2	2	0	0	1	326	0	0	1	81.75	81.75	6.353147				
15	AAEL011642	34	77	235	235	18554	27057	123	123	145.25	11464.25	78.92771	6.30246				
16	AAEL009790	20	58	177	177	14607	18748	163	163	108	8420.25	77.96528	6.28476				
17	AAEL002246	1	1	0	0	46	39	35	35	0.5	38.75	77.5	6.276124				
18	AAEL027068	2	1	1	1	3	2	173	173	1.25	87.75	70.2	6.133399				
19	AAEL012852	4	5	3	3	374	623	7	7	3.75	252.75	67.4	6.074677				
20	AAEL014617	2	0	1	1	54	90	62	62	1	67	67	6.066089				
21	AAEL005656	9	8	13	13	3	2791	10	10	10.75	703.5	65.44186	6.032142				
22	AAEL007542	80	72	76	76	7406	10240	43	43	76	4433	58.32895	5.86614				
23	AAEL004590	1	1	2	2	151	194	0	0	1.5	86.25	57.5	5.84549				
24	AAEL022419	0	0	1	1	70	7	18	18	0.5	28.25	56.5	5.820179				
25	AAEL023843	4	3	0	0	83	109	101	101	1.75	98.5	56.28571	5.814697				

Figure 9: Log₂ fold change transformation

S23																	
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O		
1	Gene_VectorBase_identifier	Bello_1	Bello_2	Bello_3	Bello_4	Cucuta_1	Cucuta_2	Cucuta_3	Cucuta_4	Mean_Be_T	Mean_Cuc_T	FC	logFC	P_Value			
2	AAEL026689	2	2	0	0	2446	3125	6	6	1	1395.75	1395.75	10.44682	0.137564			
3	AAEL020127	2	2	3	3	4652	6699	24	24	2.5	2849.75	1139.9	10.15469	0.141857			
4	AAEL007951	0	1	0	0	99	100	155	155	0.25	127.25	509	8.991522	0.000214			
5	AAEL013757	0	2	0	0	13	59	171	171	0.5	103.5	207	7.693487	0.042378			
6	AAEL005656	0	2	1	1	5	673	6	6	1	172.5	172.5	7.430453	0.34361			
7	AAEL013031	166	115	57	57	31240	36217	57	57	98.75	16892.75	171.0658	7.418408	0.136528			
9	AAEL014893	0	1	0	0	56	4	41	41	0.25	35.5	142	7.149747	0.019056			
10	AAEL026896	2	2	7	7	1021	1107	16	16	4.5	540	120	6.906891	0.12764			
11	AAEL017547	1	0	0	0	43	17	19	19	0.25	24.5	98	6.61471	0.007822			
14	AAEL005656	2	2	0	0	1	326	0	0	1	81.75	81.75	6.353147	0.3596			
15	AAEL011642	34	77	235	235	18554	27057	123	123	145.25	11464.25	78.92771	6.30246	0.145777			
16	AAEL009790	20	58	177	177	14607	18748	163	163	108	8420.25	77.96528	6.28476	0.136837			
17	AAEL002246	1	1	0	0	46	39	35	35	0.5	38.75	77.5	6.276124	6.34E-06			
18	AAEL027068	2	1	1	1	3	2	173	173	1.25	87.75	70.2	6.133399	0.129358			
19	AAEL012852	4	5	3	3	374	623	7	7	3.75	252.75	67.4	6.074677	0.149594			
20	AAEL014617	2	0	1	1	54	90	62	62	1	67	67	6.066089	0.00016			
21	AAEL005656	9	8	13	13	3	2791	10	10	10.75	703.5	65.44186	6.032142	0.357902			
22	AAEL007542	80	72	76	76	7406	10240	43	43	76	4433	58.32895	5.86614	0.144768			
23	AAEL004590	1	1	2	2	151	194	0	0	1.5	86.25	57.5	5.84549	0.144742			
24	AAEL022419	0	0	1	1	70	7	18	18	0.5	28.25	56.5	5.820179	0.097712			
25	AAEL023843	4	3	0	0	83	109	101	101	1.75	98.5	56.28571	5.814697	2.4E-06			

Figure 10: Metadata sheet after p-value calculation

Q3																				
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P				
1	Gene_VectorBase_identifier	Bello_1	Bello_2	Bello_3	Bello_4	Cucuta_1	Cucuta_2	Cucuta_3	Cucuta_4	Mean_Be_T	Mean_Cuc_T	FC	logFC	P_Value	og10_p_value					
2	AAEL026689	2	2	0	0	2446	3125	6	6	1	1395.75	1395.75	10.44682	0.137564	0.86149396					
3	AAEL020127	2	2	3	3	4652	6699	24	24	2.5	2849.75	1139.9	10.15469	0.141857	0.84814995					
4	AAEL007951	0	1	0	0	99	100	155	155	0.25	127.25	509	8.991522	0.000214	3.66890556					
5	AAEL013757	0	2	0	0	13	59	171	171	0.5	103.5	207	7.693487	0.042378	1.37286351					
6	AAEL005656	0	2	1	1	5	673	6	6	1	172.5	172.5	7.430453	0.34361	0.46393409					
7	AAEL013031	166	115	57	57	31240	36217	57	57	98.75	16892.75	171.0658	7.418408	0.136528	0.86477841					
9	AAEL014893	0	1	0	0	56	4	41	41	0.25	35.5	142	7.149747	0.019056	1.7199607					
10	AAEL026896	2	2	7	7	1021	1107	16	16	4.5	540	120	6.906891	0.12764	0.89401213					
11	AAEL017547	1	0	0	0	43	17	19	19	0.25	24.5	98	6.61471	0.007822	2.10669073					
14	AAEL005656	2	2	0	0	1	326	0	0	1	81.75	81.75	6.353147	0.3596	0.44418023					
15	AAEL011642	34	77	235	235	18554	27057	123	123	145.25	11464.25	78.92771	6.30246	0.145777	0.83631018					
16	AAEL009790	20	58	177	177	14607	18748	163	163	108	8420.25	77.96528	6.28476	0.136837	0.863795					
17	AAEL002246	1	1	0	0	46	39	35	35	0.5	38.75	77.5	6.276124	6.34E-06	5.1979893					
18	AAEL027068	2	1	1	1	3	2	173	173	1.25	87.75	70.2	6.133399	0.129358	0.88820689					
19	AAEL012852	4	5	3	3	374	623	7	7	3.75	252.75	67.4	6.074677	0.149594	0.82508461					
20	AAEL014617	2	0	1	1	54	90	62	62	1	67	67	6.066089	0.00016	3.79456346					
21	AAEL005656	9	8	13	13	3	2791	10	10	10.75	703.5	65.44186	6.032142	0.357902	0.44623628					
22	AAEL007542	80	72	76	76	7406	10240	43	43	76	4433	58.32895	5.86614	0.144768	0.83932657					
23	AAEL004590	1	1	2	2	151	194	0	0	1.5	86.25	57.5	5.84549	0.144742	0.83940465					
24	AAEL022419	0	0	1	1	70	7	18	18	0.5	28.25	56.5	5.820179	0.097712	1.0100517					
25	AAEL023843	4	3	0	0	83	109	101	101	1.75	98.5	56.28571	5.814697	2.4E-06	5.62010918					

Figure 11: $-\log_{10}(\text{p-value})$ transformation

R22																				
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q			
1	Gene_VectorBase_identifier	Bello_1	Bello_2	Bello_3	Bello_4	Cucuta_1	Cucuta_2	Cucuta_3	Cucuta_4	Mean_Be_T	Mean_Cuc_T	FC	logFC	P_Value	og10_p_value	Up/Down				
2	AAEL026689	2	2	0	0	2446	3125	6	6	1	1395.75	1395.75	10.44682	0.137564	0.86149396	UP				
3	AAEL020127	2	2	3	3	4652	6699	24	24	2.5	2849.75	1139.9	10.15469	0.141857	0.84814995	UP				
4	AAEL007951	0	1	0	0	99	100	155	155	0.25	127.25	509	8.991522	0.000214	3.66890556	UP				
5	AAEL013757	0	2	0	0	13	59	171	171	0.5	103.5	207	7.693487	0.042378	1.37286351	UP				
6	AAEL005656	0	2	1	1	5	673	6	6	1	172.5	172.5	7.430453	0.34361	0.46393409	UP				
7	AAEL013031	166	115	57	57	31240	36217	57	57	98.75	16892.75	171.0658	7.418408	0.136528	0.86477841	UP				
9	AAEL014893	0	1	0	0	56	4	41	41	0.25	35.5	142	7.149747	0.019056	1.7199607	UP				
10	AAEL026896	2	2	7	7	1021	1107	16	16	4.5	540	120	6.906891	0.12764	0.89401213	UP				
11	AAEL017547	1	0	0	0	43	17	19	19	0.25	24.5	98	6.61471	0.007822	2.10669073	UP				
14	AAEL005656	2	2	0	0	1	326	0	0	1	81.75	81.75	6.353147	0.3596	0.44418023	UP				
15	AAEL011642	34	77	235	235	18554	27057	123	123	145.25	11464.25	78.92771	6.30246	0.145777	0.83631018	UP				
16	AAEL009790	20	58	177	177	14607	18748	163	163	108	8420.25	77.96528	6.28476	0.136837	0.863795	UP				
17	AAEL002246	1	1	0	0	46	39	35	35	0.5	38.75	77.5	6.276124	6.34E-06	5.1979893	UP				
18	AAEL027068	2	1	1	1	3	2	173	173	1.25	87.75	70.2	6.133399	0.129358	0.88820689	UP				
19	AAEL012852	4	5	3	3	374	623	7	7	3.75	252.75	67.4	6.074677	0.149594	0.82508461	UP				
20	AAEL014617	2	0	1	1	54	90	62	62	1	67	67	6.066089	0.00016	3.79456346	UP				
21	AAEL005656	9	8	13	13	3	2791	10	10	10.75	703.5	65.44186	6.032142	0.357902	0.44623628	UP				
22	AAEL007542	80	72	76	76	7406	10240	43	43	76	4433	58.32895	5.86614	0.144768	0.83932657	UP				
23	AAEL004590	1	1	2	2	151	194	0	0	1.5	86.25	57.5	5.84549	0.144742	0.83940465	UP				
24	AAEL022419	0	0	1	1	70	7	18	18	0.5	28.25	56.5	5.820179	0.097712	1.0100517	UP				
25	AAEL023843	4	3	0	0	83	109	101	101	1.75	98.5	56.28571	5.814697	2.4E-06	5.62010918	UP				

Figure 12: Classification of genes as upregulated or downregulated

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Gene_VectorBase_identifier	Bello_1	Bello_2	Bello_3	Bello_4	Cucuta_1	Cucuta_2	Cucuta_3	Cucuta_4	Mean_Bello	Mean_Cucuta	FC	logFC	P_Value	log10_p_value	Up/Down	Significance	Not_Significant
2	AAEL026689	2	2	0	0	2446	3125	6	6	1	1395.75	1395.75	10.44682	0.137564	0.86149396	UP	Not significant	
3	AAEL020127	2	2	3	3	4652	6699	24	24	2.5	2849.75	1139.9	10.15469	0.141857	0.84814995	UP	Not significant	
4	AAEL007951	0	1	0	0	99	100	155	155	0.25	127.25	509	8.991522	0.000214	3.66890556	UP	Significant	
5	AAEL013757	0	2	0	0	13	59	171	171	0.5	103.5	207	7.693487	0.042378	1.37286351	UP	Significant	
6	AAEL005656	0	2	1	1	5	673	6	6	1	172.5	172.5	7.430453	0.34361	0.46393409	UP	Not significant	
7	AAEL013031	166	115	57	57	31240	36217	57	57	98.75	16892.75	171.0658	7.418408	0.136528	0.86477841	UP	Not significant	
9	AAEL014893	0	1	0	0	56	4	41	41	0.25	35.5	142	7.149747	0.019056	1.7199607	UP	Significant	
10	AAEL026896	2	2	7	7	1021	1107	16	16	4.5	540	120	6.906891	0.12764	0.89401213	UP	Not significant	
11	AAEL017547	1	0	0	0	43	17	19	19	0.25	24.5	98	6.61471	0.007822	2.10669073	UP	Significant	
14	AAEL005656	2	2	0	0	1	326	0	0	1	81.75	81.75	6.353147	0.3596	0.44418023	UP	Not significant	
15	AAEL011642	34	77	235	235	18554	27057	123	123	145.25	11464.25	78.92771	6.30246	0.145777	0.83631018	UP	Not significant	
16	AAEL009790	20	58	177	177	14607	18748	163	163	108	8420.25	77.96528	6.28476	0.136837	0.863795	UP	Not significant	
17	AAEL002246	1	1	0	0	46	39	35	35	0.5	38.75	77.5	6.276124	6.34E-06	5.1979893	UP	Significant	
18	AAEL027068	2	1	1	1	3	2	173	173	1.25	87.75	70.2	6.133399	0.129358	0.88820689	UP	Not significant	
19	AAEL012852	4	5	3	3	374	623	7	7	3.75	252.75	67.4	6.074677	0.149594	0.82508461	UP	Not significant	
20	AAEL014617	2	0	1	1	54	90	62	62	1	67	67	6.066089	0.00016	3.79456346	UP	Significant	
21	AAEL005656	9	8	13	13	3	2791	10	10	10.75	703.5	65.44186	6.032142	0.357902	0.44623628	UP	Not significant	
22	AAEL007542	80	72	76	76	7406	10240	43	43	76	4433	58.32895	5.86614	0.144768	0.83932657	UP	Not significant	
23	AAEL004590	1	1	2	2	151	194	0	0	1.5	86.25	57.5	5.84549	0.144742	0.83940465	UP	Not significant	
24	AAEL022419	0	0	1	1	70	7	18	18	0.5	28.25	56.5	5.820179	0.097712	1.0100517	UP	Not significant	
25	AAEL023843	4	3	0	0	83	109	101	101	1.75	98.5	56.28571	5.814697	2.4E-06	5.62010918	UP	Significant	

Figure 13: Identification of significantly differentially expressed genes

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Gene_VectorBase	Bello_1	Bello_2	Bello_3	Bello_4	Cucuta_1	Cucuta_2	Cucuta_3	Cucuta_4	mean_bello	mean_cucuta	FC	FClog	p_value	log10_p_value	UP/DOWN	SIG/NOTSIG
2	AAEL026689	2	2	0	0	2446	3125	6	6	1	1395.75	1395.75	10.45	0.14	0.86	Up	Not Significant
3	AAEL020127	2	2	3	3	4652	6699	24	24	2.5	2849.75	1139.9	10.15	0.14	0.85	Up	Not Significant
4	AAEL007951	0	1	0	0	99	100	155	155	0.25	127.25	509	8.99	0	3.67	Up	Significant
5	AAEL013757	0	2	0	0	13	59	171	171	0.5	103.5	207	7.69	0.04	1.37	Up	Significant
6	AAEL005656	0	2	1	1	5	673	6	6	1	172.5	172.5	7.43	0.34	0.46	Up	Not Significant
7	AAEL013031	166	115	57	57	31240	36217	57	57	98.75	16892.75	171.0658228	7.42	0.14	0.86	Up	Not Significant
8	AAEL014893	0	1	0	0	56	4	41	41	0.25	35.5	142	7.15	0.02	1.72	Up	Significant
9	AAEL026896	2	2	7	7	1021	1107	16	16	4.5	540	120	6.91	0.13	0.89	Up	Not Significant
10	AAEL017547	1	0	0	0	43	17	19	19	0.25	24.5	98	6.61	0.01	2.11	Up	Significant
11	AAEL005656	2	2	0	0	1	326	0	0	1	81.75	81.75	6.35	0.36	0.44	Up	Not Significant
12	AAEL011642	34	77	235	235	18554	27057	123	123	145.25	11464.25	78.92771084	6.3	0.15	0.84	Up	Not Significant
13	AAEL009790	20	58	177	177	14607	18748	163	163	108	8420.25	77.96527778	6.28	0.14	0.86	Up	Not Significant
14	AAEL002246	1	1	0	0	46	39	35	35	0.5	38.75	77.5	6.28	0	5.2	Up	Significant
15	AAEL027068	2	1	1	1	3	2	173	173	1.25	87.75	70.2	6.13	0.13	0.89	Up	Not Significant
16	AAEL012852	4	5	3	3	374	623	7	7	3.75	252.75	67.4	6.07	0.15	0.88	Up	Not Significant
17	AAEL014617	2	0	1	1	54	90	62	62	1	67	67	6.07	0	3.79	Up	Significant
18	AAEL005656	9	8	13	13	3	2791	10	10	10.75	703.5	65.44188047	6.03	0.36	0.45	Up	Not Significant
19	AAEL007542	80	72	76	76	7406	10240	43	43	76	4433	58.32894737	5.87	0.14	0.84	Up	Not Significant
20	AAEL004590	1	1	2	2	151	194	0	0	1.5	86.25	57.5	5.85	0.14	0.84	Up	Not Significant
21	AAEL022419	0	0	1	1	70	7	18	18	0.5	28.25	56.5	5.82	0.1	1.01	Up	Not Significant
22	AAEL023843	4	3	0	0	83	109	101	101	1.75	98.5	56.28571429	5.81	0	5.62	Up	Significant
23	AAEL010338	6	8	4	4	321	890	0	0	5.5	303.5	55.18181818	5.79	0.21	0.68	Up	Not Significant
24	AAEL000095	0	1	0	0	6	0	22	22	0.25	12.5	50	5.64	0.07	1.14	Up	Not Significant
25	AAEL021875	1	0	0	0	13	13	12	12	0.25	12.5	50	5.64	0	7.21	Up	Significant
26	AAEL005656	13	9	21	21	8	3072	16	16	16	778	48.623	5.6	0.36	0.45	Up	Not Significant
27	AAEL027468	0	1	4	4	202	228	2	2	2.25	108.5	48.22222222	5.59	0.14	0.87	Up	Not Significant
28	AAEL005656	16	16	12	12	6	2616	4	4	14	657.5	46.96428571	5.35	0.36	0.44	Up	Not Significant
29	AAEL021514	0	8	0	0	69	87	57	57	2	87.5	43.75	5.45	0	4.77	Up	Significant
30	AAEL024403	0	5	1	1	24	36	122	122	1.75	76	43.42857143	5.44	0.03	1.5	Up	Significant
31	AAEL008117	7	15	6	6	385	440	311	311	8.5	361.75	42.55882353	5.41	0	4.53	Up	Significant
32	AAEL023941	0	2	7	7	122	155	199	199	4	168.75	42.1875	5.4	0	3.91	Up	Significant

Figure 14: Final Meta data sheet

A3																		
										</								

Figure 15: Top 3 gene highlighted

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1	Gene_Vec	Bello_1	Bello_2	Bello_3	Bello_4	Cucuta_1	Cucuta_2	Cucuta_3	Cucuta_4	mean_bello	mean_cuc	FC	log2log	p_value	log10p_value				
2	AAEL0266	2	2	0	0	2446	3125	6	6	1	1395.75	1395.75	10.45	0.14	0.86				
3	AAEL0201	2	2	3	3	4652	6699	24	24	2.5	2849.75	1139.9	10.15	0.14	0.85				
4	AAEL0079	0	1	0	0	99	100	155	155	0.25	127.25	509	8.99	0	3.67				
5	AAEL0137	0	2	0	0	13	59	171	171	0.5	103.5	207	7.69	0.04	1.37				
6	AAEL0056	0	2	1	1	5	673	6	6	1	172.5	172.5	7.43	0.34	0.46				
7	AAEL0130	166	115	57	57	31240	36217	57	57	98.75	16892.75	171.0658	7.42	0.14	0.86				
8	AAEL0148	0	1	0	0	56	4	41	41	0.25	35.5	142	7.15	0.02	1.72				
9	AAEL0268	2	2	7	7	1021	1107	16	16	4.5	540	120	6.91	0.13	0.89				
10	AAEL0175	1	0	0	0	43	17	19	19	0.25	24.5	98	6.61	0.01	2.11				
11	AAEL0056	2	2	0	0	1	326	0	0	1	81.75	81.75	6.35	0.36	0.44				
12	AAEL0116	34	77	235	235	18554	27057	123	123	145.25	11464.25	78.92771	6.3	0.15	0.84				
13	AAEL0097	20	58	177	177	14607	18748	163	163	108	8420.25	77.96528	6.28	0.14	0.86				
14	AAEL0022	1	1	0	0	46	39	35	35	0.5	38.75	77.5	6.28	0	5.2				
15	AAEL0270	2	1	1	1	3	2	173	173	1.25	87.75	70.2	6.13	0.13	0.89				
16	AAEL0128	4	5	3	3	374	623	7	7	3.75	252.75	67.4	6.07	0.15	0.83				
17	AAEL0146	2	0	1	1	54	90	62	62	1	67	67	6.07	0	3.79				
18	AAEL0056	9	8	13	13	3	2791	10	10	10.75	703.5	65.44186	6.03	0.36	0.45				
19	AAEL0075	80	72	76	76	7406	10240	43	43	76	4433	58.32895	5.87	0.14	0.84				
20	AAEL0045	1	1	2	2	151	194	0	0	1.5	86.25	57.5	5.85	0.14	0.84				
21	AAEL0224	0	0	1	1	70	7	18	18	0.5	28.25	56.5	5.82	0.1	1.01				

Figure 16: Volcano plot data sheet

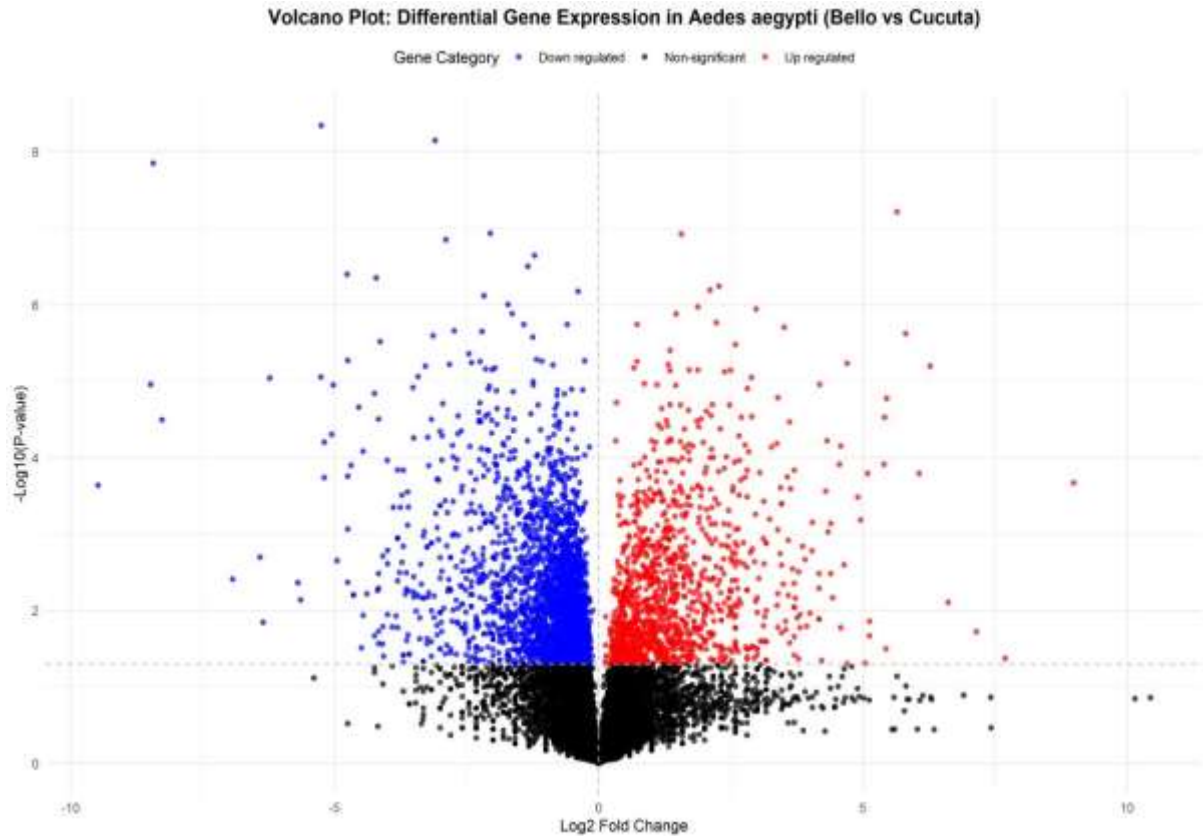


Figure 17: Volcano plot of *Aedes aegypti* Genes in Bello and Cucuta populations

The volcano plot shows differential gene expression between *Aedes aegypti* populations from Bello and Cúcuta. The x-axis represents log₂ fold change, with positive values indicating upregulation in Bello and negative values indicating upregulation in Cúcuta. The y-axis shows $-\log_{10}(\text{p-value})$, with higher values indicating greater statistical significance. Genes with $|\log_2\text{FC}| > 1$ and $p < 0.05$ are considered differentially expressed. Significantly **up-regulated** genes of the resistant line are denoted by red colour and the **down-regulated** ones by blue and the **non-significant** ones by black. This visualization highlights genes with large expression differences and strong statistical significance.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Gene_Vector	Bello_1	Bello_2	Bello_3	Bello_4	Cucuta_1	Cucuta_2	Cucuta_3	Cucuta_4	mean_bello	mean_cuc	FC	FC2log	p_value	log2Up vs UP/DOWN	SIG/NOTSIG						
2	AAEL007951	0	1	0	0	99	300	355	355	0.25	127.25	509	8.99	0	3.67 Up	Significant						
3	AAEL013757	0	1	0	0	13	59	171	171	0.5	101.5	207	7.88	0.04	1.37 Up	Significant						
4	AAEL014893	0	1	0	0	56	4	41	41	0.25	35.5	142	7.15	0.02	1.72 Up	Significant						
5	AAEL017547	1	0	0	0	43	17	19	19	0.25	24.5	98	6.81	0.01	2.11 Up	Significant						
6	AAEL002246	1	1	0	0	46	39	35	35	0.5	38.75	77.5	6.28	0	5.2 Up	Significant						
7	AAEL014617	2	0	1	1	54	90	62	62	1	67	67	6.07	0	3.79 Up	Significant						
8	AAEL023843	4	3	0	0	83	309	101	101	1.75	98.5	56.28571	5.81	0	5.62 Up	Significant						
9	AAEL021875	1	0	0	0	13	13	12	12	0.25	12.5	50	5.64	0	7.21 Up	Significant						
10	AAEL021514	0	8	0	0	69	87	97	97	2	87.5	43.75	5.45	0	4.77 Up	Significant						
11	AAEL024403	0	5	1	1	24	36	122	122	1.75	76	43.42857	5.44	0.03	1.5 Up	Significant						
12																						
13																						
14																						
15																						
16																						
17																						

Figure 18: Heat map data sheet

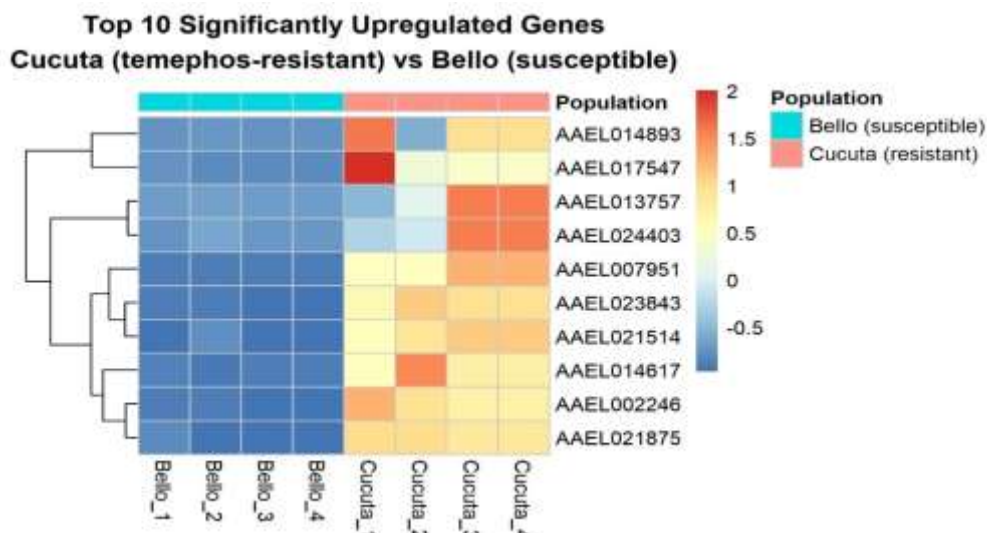


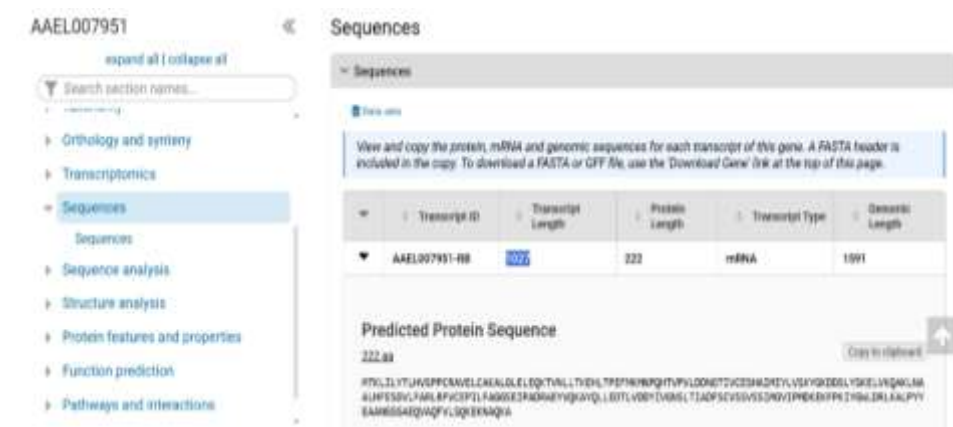
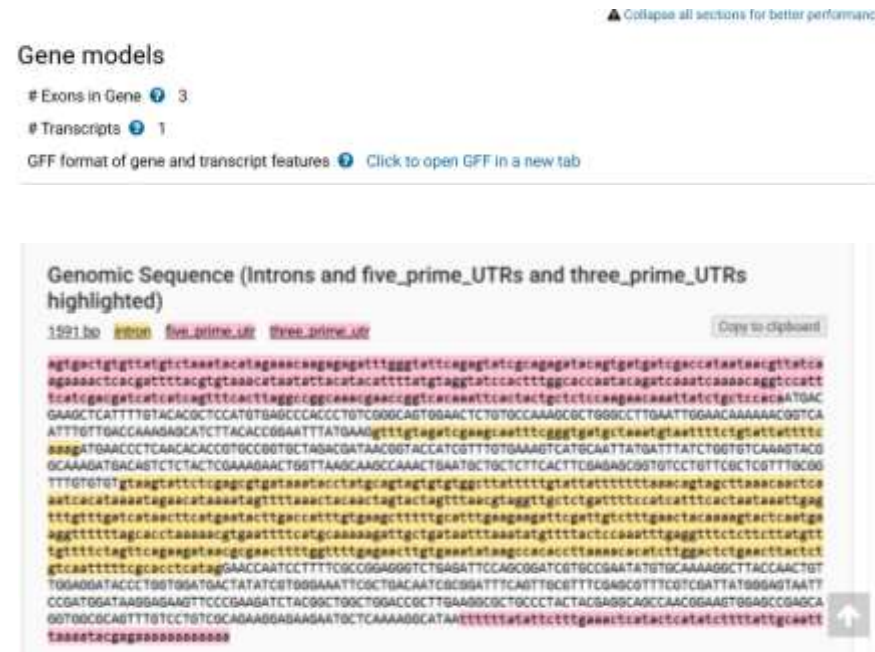
Figure 19: Heat map of DEGs in Bello and Cucuta populations of *Aedes aegypti*

The heat map displays Z-score-normalized expression of differentially expressed genes between *Aedes aegypti* populations from Bello and Cúcuta. Genes are on the y-axis, and biological replicates on the x-axis. The color gradient (blue to red) indicates lower to higher gene expression. Bello samples show lower expression, while Cúcuta samples show higher expression. Hierarchical clustering confirms strong population grouping, supporting the upregulation in Cúcuta identified in the DESeq2 analysis, confirming consistent overexpression in Cúcuta.

Results of Top 3 genes

The top three genes identified as overexpressed in the Cucuta population are,

[AAEL007951](#), [AAEL013757](#), [AAEL014893](#)

Transcript Vector Base Identifier	AAEL007951
Size of gene	 1027 basepairs
Number of introns/ exons	Exons 02 & Introns 02  Highlighted area in yellow color is the intron.

Chromosomal location and position on chromosome

AAEL007951 glutathione transferase

Name	GSTE2
Gene Type	protein coding gene
Biotype Classification	protein_coding
Chromosome	2
Location	Aaegl5_2:351,681,351-351,682,941(+)
Species	<i>Aedes aegypti</i>
Strain	LVP_AGWG Data set
Status	Reference Strain
User Comments	Add the first
Community Annotations	View / Update in Apollo editor

Genomic Location

Gene Location				
Download Data sets				
Sequence Level	Genome Sequence	Location	organismAbbrev	Genome Browser
chromosome	Aaegl5_2	351681351 - 351682941 (+)	aaeglVP_AGWG	View

Size of genomic DNA

Sequences					
Data sets					
View and copy the protein, mRNA and genomic sequences for each transcript of this gene. A FASTA header is included in the copy. To download a FASTA or GFF file, use the 'Download Gene' link at the top of this page.					
Transcript ID	Transcript Length	Protein Length	Transcript Type	Genomic Length	
AAEL007951-RB	1027	222	mRNA	1591	

1591 basepairs

The protein sequence

Predicted Protein Sequence

222 aa

[Copy to clipboard](#)

```
MTKLILYTLHVSPPCRAVELCAKALGLELEQKTVNLLTKEHLTPEFMKMNPHQHTVPVLDNGTIVCESHAIMIYLVSKYGGKDDSLYSKELVKQAKLNA
ALHFESGVLFAFLRFVCEPIIFAGGSEIPADRAEYVQKAYQLLEDTLVDDYIVGNSLTIADFSVCVSSVSSIMGVIPMDKEKFKIYGWLDRLKALPY
EAANGSGAEQVAQFVLSQKEKNAQKA
```

> AAEL007951

```
MTKLILYTLHVSPPCRAVELCAKALGLELEQKTVNLLTKEHLTPEFMKMNPHQHTVPVLD
DNGTIVCESHAIMIYLVSKYGGKDDSLYSKELVKQAKLNAALHFESGVLFAFLRFVCEPI
LFAGGSEIPADRAEYVQKAYQLLEDTLVDDYIVGNSLTIADFSVCVSSVSSIMGVIPMDK
EKFPKIYGWLDRLKALPYEAANGSGAEQVAQFVLSQKEKNAQKA
```

Orthologs in non-*Aedes* mosquito species and number of orthologs found

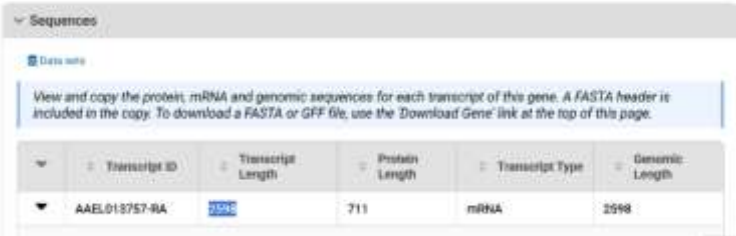
Search this table...							
1,408 rows							
Cluster Omega	Transcript	Product	Organism	Protein Length	Reference Strain?	Systemic?	has comments
☐	AAEL001059-RA	glutathione S-transferase (GSTD3)	<i>Aedes aegypti</i> LVP_AGWG	215	yes	no	no
☐	AAEL001061-RB	glutathione S-transferase (GSTD1)	<i>Aedes aegypti</i> LVP_AGWG	209	yes	no	no
☐	AAEL001061-RC	glutathione S-transferase (GSTD1)	<i>Aedes aegypti</i> LVP_AGWG	211	yes	no	no
☐	AAEL001061-RA	glutathione S-transferase (GSTD1)	<i>Aedes aegypti</i> LVP_AGWG	216	yes	no	no
☐	AAEL007948-RA	glutathione transferase	<i>Aedes aegypti</i> LVP_AGWG	223	yes	no	no
☐	AAEL007954-RB	glutathione transferase	<i>Aedes aegypti</i> LVP_AGWG	224	yes	no	no
☐	AAEL007962-RA	glutathione transferase	<i>Aedes aegypti</i> LVP_AGWG	224	yes	no	no

1408 orthologs identified for both *Aedes* and non-*Aedes* species. Only 1389 orthologs identified for non-*Aedes* mosquito species.

Show only one transcript per gene							
Search this table...							
1,408 rows							
Keep checked values at top							
1389 genes selected. Clear selection.							
Cluster Omega	Transcript	Product	Organism	Protein Length	Reference Strain?	Systemic?	has comments
☐	AAEL001059-RA	glutathione S-transferase (GSTD3)	<i>Aedes aegypti</i> LVP_AGWG	215	yes	no	no
☐	AAEL001061-RB	glutathione S-transferase (GSTD1)	<i>Aedes aegypti</i> LVP_AGWG	209	yes	no	no
☐	AAEL001061-RC	glutathione S-transferase (GSTD1)	<i>Aedes aegypti</i> LVP_AGWG	211	yes	no	no
☐	AAEL001061-RA	glutathione S-transferase (GSTD1)	<i>Aedes aegypti</i> LVP_AGWG	216	yes	no	no
☐	AAEL007948-RA	glutathione transferase	<i>Aedes aegypti</i> LVP_AGWG	223	yes	no	no
☐	AAEL007954-RB	glutathione transferase	<i>Aedes aegypti</i> LVP_AGWG	224	yes	no	no
☐	AAEL007962-RA	glutathione transferase	<i>Aedes aegypti</i> LVP_AGWG	224	yes	no	no
☐	AAEL009016-RB	glutathione transferase	<i>Aedes aegypti</i> LVP_AGWG	227	yes	no	no
☐	AAEL009017-RA	glutathione transferase	<i>Aedes aegypti</i> LVP_AGWG	229	yes	no	no

Methods for assigning predicting function to the gene

My Organism Preferences (88 of 88) enabled			
EC Numbers			
Download Data sets			
Warning: some EC numbers are computationally inferred by orthology. Examine the evidence in the 'EC Numbers (scores for computational inference)' table below. Also, explore this gene's ortholog group at OrthoMCL: OG6_108130			
EC Description	Assignment Source	EC Number	Transcript ID(s)
Glutathione transferase	computationally inferred from orthology	2.5.1.18	AAEL007961-RB

Transcript Vector Base Identifier Size of gene	AAEL013757  2598 basepairs
Number of introns/ exons	Exons 02 & Introns 0  No introns shown

Chromosomal location and position on chromosome

AAEL013757 hexamerin 2 beta

Gene Type

Biotype Classification

Chromosome

Location

Species

Strain

Status

User Comments

Community Annotations

Model Organism Database(s)

FlyBase

protein coding gene

protein_coding

1

Aaegl_5.1 276,878,506-276,881,103 (+)

Aedes aegypti

LVP_AQWG

Reference Strain

Add the first

View / Update in Apollo editor

Fbgn0002562, Fbgn0002563, Fbgn0002564, Fbgn0002565

Genomic Location

Gene Location

Download

Data sets

Sequence Level	Genome Sequence	Location	organismAbbrev	Genome Browser
chromosome	Aaegl_5.1	276878506-276881103 (*)	aaeglLVP_AQWG	View

Size of genomic DNA

My Organism Preferences (88 of 88) enabled

Sequences

Data sets

View and copy the protein, mRNA and genomic sequences for each transcript of this gene. A FASTA header is included in the copy. To download a FASTA or GFF file, use the 'Download Gene' link at the top of this page.

Transcript ID	Transcript Length	Protein Length	Transcript Type	Genomic Length
AAEL013757-RA	2598	711	mRNA	2598

2598 basepairs

The protein sequences

Predicted Protein Sequence

711 aa

Copy to clipboard

MRSLAVTVVCLFALASGVHTKQWYQDEIFLXQKASLEIFQKHQPELHTSSMEESKQFDIQYLHDTNARVKEFLAFYVHDLPLKEEFSIQ
NEFHREEVZALFHLFYAKQMDTFYKTHNARPHNSPHFYALTVAILHMHQREZVLPAFYEIYFFPQDNVTSKAGRYXQGFYQKXCINDVYT
AYIPSAHTSYVHTGPEQWVYFTSDIGLNSYYYFPAQVYFMNGKEFGLYKDRROEFFLPQHQQLARHYLERLNDLSVIPPSSAKSITTEYYP
YLRVYNGZPPFVRQNYYSASVDKSYVVEGIEDFHRIDREAZIYHFIIVLPQKYVNZITTPSGEYV,GNLIQANDSVNTRFYVLESLAKYFLOOSFEL
QHEYKZPSVLEKFETSHQPHFYQYKIVIEVYVYRFLDLLPSYKKEELOFPQVIZETVDEKLVITYFDKFDAGITNAQDVVEFDETINKSSQWKFQ
KZAHYQGSDFVIYARPHLNLPTFKLNVVSEKAKATVYVYLSPKYDQVQKIVQGDARENFFELDFLVDLVPGNAITRNSQDFSMFYDITTY
FELVQZNNKAVTSGKFFLPQSEAKCIPKRLHLKQKXGGFFQFFHVPVWVWVEQFTGPDHTLSCQVSGARYLDLPFGYFFERKIDENAF
TRUNKYVDTLIYHSETEVIDVSDH

>AAEL013757

MRSILAVTVVCLFALASGSYMTKDVKYGDEIFLKKQKAILEIFQHIHQPELHTSSWEESKG
FDIEQYLDHYTNAEAVKEFLRFYKHGMLPLEEIFSIQNEFHREEVIALFHLFYYAKDWDTF
YKTMVWARFHVNEGFMFVYALTVAVLHRHDMEGIVLPAPYEIYPFYFFNDVVISKAQRYKMQ
GFYKMKKINDVYTAYIPSNYTGYYVHTGPEQRVGYFTEDIGLNSYYYYFHADYPFWMGGKE
FGLYKDRRGFFLFQHQQLLARYYLERLSNDLGVIPFWSHKSITTGYYPYLRYNGIPFP
VRDNYYSASVDKSYVVEQIEDFEHRIMEAIDYGFIVLPDGKYVNITTPSGIEYLGNIQSN
GDSVNTRFYGYLEILAKYFLGGSFKLQHEYRAIPSVLEKFETSMRDPMFYQFYKRIVEYYY
RFLDLLPSYKKEELDFPGVKIETVEMDKLVTYFDKFDADITNAVDVEVFDETTMKGSMDKK
FGKIAHYQGEDFVIYARMPRLNHLPTFTKLNVSSEKAQKATVFVYLGPKYDQYGNIIYGVDA
NRENFELDHLVDLVPKNAITRNSQDFSFWVKDRTTYFELYQQVMKAVTGEGKFPDMS
EAHCGFPARLMLPKGKKGMPFQFFFMVVPYHAPKVEQFTGFDHTLSCGVGSGARYLDDL
FGYPFERKIDENAWFTPNMKYYDTLIYHKSETEVNSVVIH

**Orthologs in
non- Aedes
mosquito species
and number of
orthologs found**

My Organism Preferences (88 of 88) ☒ enabled

Search this table...

Clustal Omega	Transcript	Product	Organism	Protein Length	Reference Strain?	Synthetic?	has comments
☐	AAEL008045-RA	hexamerin 2 beta	Aedes aegypti LVP_AGWG	692	yes	no	no
☐	AAEL008817-RA	hexamerin 2 beta	Aedes aegypti LVP_AGWG	708	yes	no	no
☐	AAEL011169-RA	hexamerin 2 beta	Aedes aegypti LVP_AGWG	712	yes	no	no
☐	AAEL013861-RA	hexamerin 2 beta	Aedes aegypti LVP_AGWG	692	yes	no	no
☐	AAEL013863-RB	hexamerin 2 beta	Aedes aegypti LVP_AGWG	692	yes	no	no
☐	AAEL013990-RB	hexamerin 2 beta	Aedes aegypti LVP_AGWG	691	yes	no	no
☐	AAEL021888-RA	unspecified product	Aedes aegypti	692	yes	no	no

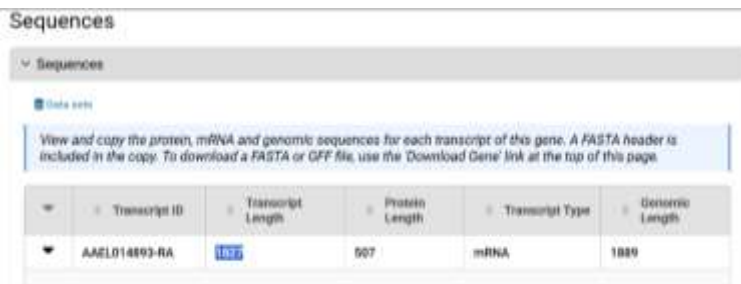
459 orthologs identified for both Aedes and non-Aedes species. Only 6 orthologs identified for non-Aedes mosquito species.

☐ Show only one transcript per gene

Search this table... 459 rows ☒ Keep checked values at top 5 genes selected. Clear selection.

Clustal Omega	Transcript	Product	Organism	Protein Length	Reference Strain?	Synthetic?	
☐	AAEL008045-RA	hexamerin 2 beta	Aedes aegypti LVP_AGWG	692	yes	no	no
☐	AAEL008817-RA	hexamerin 2 beta	Aedes aegypti LVP_AGWG	708	yes	no	no
☐	AAEL011169-RA	hexamerin 2 beta	Aedes aegypti LVP_AGWG	712	yes	no	no
☐	AAEL013861-RA	hexamerin 2 beta	Aedes aegypti LVP_AGWG	692	yes	no	no
☐	AAEL013863-RB	hexamerin 2 beta	Aedes aegypti LVP_AGWG	692	yes	no	no
☐	AAEL013990-RB	hexamerin 2 beta	Aedes aegypti LVP_AGWG	691	yes	no	no
☐	AAEL021888-RA	unspecified product	Aedes aegypti LVP_AGWG	692	yes	no	no
☐	AAEL036_006095.R0488	hexamerin-1.1-like	Aedes albopictus C6/36 cell line	691	no	no	no

Methods for assigning predicting function to the gene	Function prediction EC Numbers  Data sets No data available EC Numbers (scores for computational inference)  Data sets No data available GO Slim  Data sets No data available GO Terms  Data sets No data available
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Transcript Vector Base Identifier Size of gene	AAEL014893  1827 basepairs																				
Number of introns/ exons	Exons 02 & Introns 01  Highlighted area in yellow color is the intron.																				
Chromosomal location and position on chromosome	AAEL014893 cytochrome P450 <table border="1"> <tr> <td>Name</td> <td>CYP6B82</td> </tr> <tr> <td>Gene Type</td> <td>protein coding gene</td> </tr> <tr> <td>Biotype Classification</td> <td>protein_coding</td> </tr> <tr> <td>Chromosome</td> <td>1</td> </tr> <tr> <td>Location</td> <td>Aaegl.5:1,371,328,495-271,330,383</td> </tr> <tr> <td>Species</td> <td>Aedes aegypti</td> </tr> <tr> <td>Strain</td> <td>LVP_AGWC Data set</td> </tr> <tr> <td>Status</td> <td>Reference Strain</td> </tr> <tr> <td>User Comments</td> <td>Add the first</td> </tr> <tr> <td>Community Annotations</td> <td>View / Update in Apollo editor</td> </tr> </table>	Name	CYP6B82	Gene Type	protein coding gene	Biotype Classification	protein_coding	Chromosome	1	Location	Aaegl.5:1,371,328,495-271,330,383	Species	Aedes aegypti	Strain	LVP_AGWC Data set	Status	Reference Strain	User Comments	Add the first	Community Annotations	View / Update in Apollo editor
Name	CYP6B82																				
Gene Type	protein coding gene																				
Biotype Classification	protein_coding																				
Chromosome	1																				
Location	Aaegl.5:1,371,328,495-271,330,383																				
Species	Aedes aegypti																				
Strain	LVP_AGWC Data set																				
Status	Reference Strain																				
User Comments	Add the first																				
Community Annotations	View / Update in Apollo editor																				

	Genomic Location Gene Location Download Gene sets <table><thead><tr><th>Sequence Level</th><th>Genomic Sequence</th><th>Location</th><th>organismAbbrev</th><th>Genomic Browser</th></tr></thead><tbody><tr><td>chromosome</td><td>Aae01.1</td><td>271328495 - 271330383 (3)</td><td>ae01.VP_A01W0</td><td>View</td></tr></tbody></table>	Sequence Level	Genomic Sequence	Location	organismAbbrev	Genomic Browser	chromosome	Aae01.1	271328495 - 271330383 (3)	ae01.VP_A01W0	View
Sequence Level	Genomic Sequence	Location	organismAbbrev	Genomic Browser							
chromosome	Aae01.1	271328495 - 271330383 (3)	ae01.VP_A01W0	View							
Size of genomic DNA	Sequences Data sets View and copy the protein, mRNA and genomic sequences for each transcript of this gene. A FASTA header is included in the copy. To download a FASTA or GFF file, use the 'Download Gene' link at the top of this page. <table><thead><tr><th>Transcript ID</th><th>Transcript Length</th><th>Protein Length</th><th>Transcript Type</th><th>Genomic Length</th></tr></thead><tbody><tr><td>▼ AAEL014893-RA</td><td>1827</td><td>507</td><td>mRNA</td><td>1889</td></tr></tbody></table>	Transcript ID	Transcript Length	Protein Length	Transcript Type	Genomic Length	▼ AAEL014893-RA	1827	507	mRNA	1889
Transcript ID	Transcript Length	Protein Length	Transcript Type	Genomic Length							
▼ AAEL014893-RA	1827	507	mRNA	1889							
The protein sequences	Predicted Protein Sequence 507 aa Copy to clipboard MDPVTIVILTIFVGLTGLVYFFLRREQQKWPRLGVPFAKNPHLLFGNVRGIFQKEHSCEILQRLYWEFKGRGLKLGGINNFFQPAVLVIDPEISKILVKDFNKFHDRGIFVDPAGDPLSANLFSLEGAQWKAMRTKMSPTFTSGKMKYMFESVLNVAERLKYLAENCLKEDIELKNILQRFMTDVI GNVAFGVECNISIKNPSSSEFRLMGLKANRFDGVRFLKFFIGGAYKNFAKKIKLKVVEDDVHKFFMSLVHSTVHYREGNNVKRNDFLNLLMEIKNKGKFSDEPNSSGGEITMNEIAAQCFIFFTAGFETSSTTINFCLYELANNPDIQDRLRNEIEDVVAKDGGELKYDTLLGMNYLDRVVSETLRKYSVDNLFRISNSPYTPDGCNFTIPAGTLFQIPIHSMHHDPEYFPDGRFDPDRFLPEVAKSRHPYCYLPFGEGRVCI GMRFGMLMQTKIGLVTL LRNFRFGPRSETPDRLQFEAKTFVLTPQTGIYLIKIEPIGI										

Orthologs in non- Aedes mosquito species and number of orthologs found

Cluster Group	Transcript	Product	Organism	Protein Length	Reference Strain?	Symmetric?	Has annotation?
☐	AAEL000065-RA	Cytochrome P450	Aedes aegypti (VP_A00065)	525	yes	no	no
☐	AAEL000065-RC	Cytochrome P450	Aedes aegypti (VP_A00065)	521	yes	no	no
☐	AAEL000065-RE	Cytochrome P450	Aedes aegypti (VP_A00065)	521	yes	no	no
☐	AAEL000065-RA	Cytochrome P450	Aedes aegypti (VP_A00065)	521	yes	no	no
☐	AAEL000065-RA	Cytochrome P450 (SwissProt: P07061, Acc: A0A151F044)	Aedes aegypti (VP_A00065)	523	yes	no	no
☐	AAEL000065-RE	Cytochrome P450	Aedes aegypti (VP_A00065)	522	yes	no	no
☐	AAEL000148-RA	Cytochrome P450	Aedes aegypti	526	yes	no	no

3879 orthologs identified for both Aedes and non-Aedes species.

Methods for assigning predicting function to the gene

My Organism Preferences (88 of 88) enable			
EC Description	Assignment Source	EC Number	Transcript ID(s)
Transferred entry: 1.14.19.62	computationally inferred from orthology	1.3.3.9	AAEL014893-RA
Alpha-amylase	computationally inferred from orthology	3.2.1.1	AAEL014893-RA
With NADH or NADPH as one donor, and incorporation of one atom of oxygen.	computationally inferred from orthology	1.14.13.-	AAEL014893-RA
With reduced flavin or flavoprotein as one donor, and incorporation of one atom of oxygen.	computationally inferred from orthology	1.14.14.-	AAEL014893-RA
Unspecific monooxygenase	computationally inferred from orthology	1.14.14.1	AAEL014893-RA
Cholesterol 24-hydroxylase	computationally inferred from orthology	1.14.14.25	AAEL014893-RA
Carotene epsilon-hydroxylase	computationally inferred from orthology	1.14.14.158	AAEL014893-RA

EC Numbers (scores for computational inference)							
Download Data view							
This gene's protein sequence was compared to the protein sequences of other genes in the same ortholog group. The gene is inferred to receive a specific EC number based on how well the sequence matches genes that have been previously assigned this EC number.							
EC Number	Length Score (1-6)	BLAST Score (1-4)	Conservation Score (1-6)	# proteins with EC	# proteins in ortholog group	# genes with EC	# genes in ortholog group
1.14.13.-	4	1	4	8	9/107	8	238
1.14.14.1	4	1	4	35	9/107	12	238
1.14.14.25	4	2	4	3	9/107	9	238
1.14.14.158	4	1	4	1	9/107	1	238
3.2.1.1	4	1	4	1	9/107	1	238
1.3.3.9	4	1	4	15	9/107	1	238
1.14.14.-	4	4	4	48	9/107	19	238

Previous studies on DGE

GO Slim						
Download Data sets						
Search this table						
Ontology	GO ID	Is Not	GO Term Name	GO Slim ID	GO Slim Term Name	
Biological Process	00000114	is a	obsolete oxidation-reduction process	is a	is a	
Cellular Component	00000020	is a	membrane	00000075	cellular_component	
Cellular Component	00000021	is a	integral component of membrane	00000075	cellular_component	
Molecular Function	00000007	is a	monooxygenase activity	00000049	oxidoreductase activity	
Molecular Function	00000008	is a	iron ion binding	00000047	ion binding	
Molecular Function	00000009	is a	oxidoreductase activity	00000049	oxidoreductase activity	
Molecular Function	00000009	is a	oxidoreductase activity, acting on paired donors, with incorporation or reduction of molecular oxygen	00000049	oxidoreductase activity	
Molecular Function	00000007	is a	heme binding	00000049	molecular_function	
Molecular Function	00000007	is a	metal ion binding	00000047	ion binding	

GO Terms								
Download Data sets								
Search this table								
Ontology	GO ID	Is Not	GO Term Name	Species	Evidence Code	Reference	Support for Evidence Code Assignment	Transcript ID(s)
Biological Process	00000114	is a	obsolete oxidation-reduction process	gou_import	EA	is a	is a	AMEL014893-RA
Biological Process	00000114	is a	obsolete oxidation-reduction process	interprogo	EA	is a	is a	AMEL014893-RA
Cellular Component	00000020	is a	membrane	gou_import	EA	is a	is a	AMEL014893-RA
Cellular Component	00000021	is a	integral component of membrane	gou_import	EA	is a	is a	AMEL014893-RA
Cellular Component	00000021	is a	integral component of membrane	gou_import	EA	is a	is a	AMEL014893-RA
Molecular Function	00000007	is a	monooxygenase activity	gou_import	EA	is a	is a	AMEL014893-RA
Molecular Function	00000007	is a	monooxygenase activity	gou_import	EA	is a	is a	AMEL014893-RA
Molecular Function	00000007	is a	monooxygenase activity	gou_import	EA	is a	is a	AMEL014893-RA
Molecular Function	00000008	is a	iron ion binding	gou_import	EA	is a	is a	AMEL014893-RA
Molecular Function	00000008	is a	iron ion binding	gou_import	EA	is a	is a	AMEL014893-RA
Molecular Function	00000009	is a	oxidoreductase activity	gou_import	EA	is a	is a	AMEL014893-RA
Molecular Function	00000009	is a	oxidoreductase activity, acting on paired donors, with incorporation or reduction of molecular oxygen	gou_import	EA	is a	is a	AMEL014893-RA
Molecular Function	00000009	is a	oxidoreductase activity, acting on paired donors, with incorporation or reduction of molecular oxygen	gou_import	EA	is a	is a	AMEL014893-RA
Molecular Function	00000007	is a	heme binding	gou_import	EA	is a	is a	AMEL014893-RA
Molecular Function	00000007	is a	heme binding	gou_import	EA	is a	is a	AMEL014893-RA
Molecular Function	00000007	is a	metal ion binding	gou_import	EA	is a	is a	AMEL014893-RA

Literature

PubMed			
Download Data sets			
Shows are references from the NCBJ gene2pubmed database and from manual assignment by curators.			
PubMed ID	DOI	Title	Authors
22720108	10.1371/journal.pntd.0001692	Dene amplification, ABC transporters and cytochrome P450c: unraveling the molecular basis of pyrethroid resistance in the dengue vector, <i>Aedes aegypti</i> .	Boriami V, Jones CM, Poupardin R, Vantas J, Ranson H

DISCUSSION

Main Findings of the Study:

A key finding from this study is that the Cúcuta population of *Aedes aegypti*, resistant to temephos, shows strong upregulation of specific genes, unlike the Bello population, where genes are suppressed. The volcano plot revealed many overexpressed genes in Cúcuta, with positive log₂ fold changes indicating a strong transcriptional response. The heatmap confirmed consistent overexpression across all biological replicates, highlighting stable transcriptional changes. These findings suggest that the resistant phenotype in Cúcuta is driven by gene overexpression involved in detoxification and stress responses, showing that transcriptional adaptation plays a critical role in resistance to temephos, likely aiding larvae survival under insecticide exposure.

Link Between Statistical Analysis and Biological Interpretation:

The statistical approach of using fold change > 2 and $p < 0.05$ ensured that only genes with significant and biologically meaningful expression differences were selected. This helped prioritize genes showing substantial upregulation, aiding biological interpretation. The top three genes identified ([AAEL007951](#), [AAEL013757](#), [AAEL014893](#)) showed the highest fold changes and statistical support. Their overexpression in the Cúcuta population is linked to resistance, likely contributing to survival under temephos exposure. The gene structures (with introns/exons) and chromosomal locations are key to understanding their regulation. Notably, [AAEL007951](#) and [AAEL014893](#) contain introns, suggesting complex regulatory mechanisms that could influence their expression in response to temephos. By connecting these statistical results to gene functions, we establish a clear link between overexpression and insecticide resistance, showing that transcriptional adaptation helps mosquitoes detoxify temephos or mitigate stress.

Selection and Analysis of the Top Three Genes:

The genes [AAEL007951](#), [AAEL013757](#) and [AAEL014893](#) were selected for their high fold changes and statistical significance, confirming their biological relevance to temephos resistance. [AAEL007951](#) and [AAEL014893](#), which contain introns, suggest complex regulation, likely involving transcription factors activated under stress, potentially linked to detoxification mechanisms. In contrast, [AAEL013757](#), which is intronless, may be rapidly transcribed in response to environmental pressures like insecticide exposure. Additionally, these genes are conserved across mosquito species, including non-*Aedes* species, indicating their involvement in critical processes such as metabolism and detoxification, essential for resistance (Vontas et al., 2012). Their consistent overexpression in the Cúcuta population supports the role of transcriptional adaptation in facilitating temephos resistance.

Comparison with Existing Research:

While the functional characterization of [AAEL007951](#), [AAEL013757](#) and [AAEL014893](#) in temephos resistance is limited, many studies have shown that detoxification enzymes like cytochrome P450 monooxygenases, carboxylesterases, and glutathione S-transferases are often overexpressed in insecticide-resistant mosquitoes (Vontas et al., 2012; Hernandez et al., 2021). This study aligns with those findings, showing that upregulated genes in the Cúcuta population

resemble detoxification genes linked to resistance. This supports the idea that these genes could help detoxify temephos, reducing its efficacy. The connection between our results and previous studies highlights the biological relevance of these genes and suggests they are important targets for future research on insecticide resistance.

Wider Context of Insecticide Resistance in Mosquitoes:

Insecticide resistance in mosquitoes is a complex, multifactorial issue. This study underscores the role of transcriptional reprogramming in resistance, aligning with previous research on *Aedes aegypti* (Ranson et al., 2011). Resistance often results from the overexpression of genes related to detoxification, metabolism, and stress tolerance, rather than from single mutations. The upregulation of these genes in the Cúcuta population reinforces the idea that resistance is driven by coordinated gene expression, not just a few mutations.

These findings are critical for vector control strategies, especially in areas still using temephos. Since larvae surviving temephos exposure can mature into adults, understanding and targeting the genetic and transcriptional factors behind resistance is vital for maintaining the effectiveness of larvicides in mosquito control efforts.

Overview of the *Aedes aegypti* Genome and Transcriptome:

The genome of *Aedes aegypti* is large (~1.38 Gb) and highly repetitive, with over 15,000 protein-coding genes. The flexible transcriptome of *Aedes aegypti* allows for rapid adaptation to environmental stresses, such as insecticide exposure (Nene et al., 2007). This transcriptional plasticity is a key factor in the mosquito's ability to evolve resistance mechanisms. This study findings illustrate how the transcriptional changes observed at the larval stage, when the mosquito is exposed to temephos, directly contribute to the persistence of resistant populations.

Future Directions and My conclude Thoughts:

Future research should validate the role of specific genes in resistance mechanisms through gene silencing or overexpression, and explore the regulatory networks controlling these genes. Expanding studies to other insecticide-resistant populations will help determine whether these genes are consistently involved across regions. This study provides insights into the molecular mechanisms of insecticide resistance in *Aedes aegypti*, a key vector for diseases like dengue and Zika. It highlights the complexity of resistance, suggesting that multiple genes are involved. The findings offer exciting potential for gene-based control methods, which could be more sustainable than traditional chemical approaches. Additionally, this opens avenues for biological control agents and environmental interventions. Future research in functional genomics and epigenetics could reveal how *Aedes aegypti* adapts so quickly to human interventions, ultimately aiding in better vector control strategies and improving public health.

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